

BILLINGS LOGAN, MT, USA

WMO: 726770

Lat: 45.807N

Lon: 108.542W

Elev: 1092

StdP: 88.88

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24033

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.5	-19.3	-25.6	0.4	-21.9	-22.6	0.6	-18.6	14.7	-0.7	13.2	0.4	3.5	30	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	34.9	17.0	33.1	16.6	31.3	16.3	19.1	29.5	18.1	28.9	17.4	28.3	4.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.8	12.8	22.3	14.5	11.8	21.5	13.5	11.0	21.0	58.6	29.3	55.4	28.6	52.8	28.2	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	11.2	10.0	DB	-26.2	38.1	3.4	1.5	-28.6	39.2	-30.6	40.1	-32.5	40.9	-35.0	42.0	(4)
(5)			WB	-26.3	20.9	3.1	1.4	-28.5	21.9	-30.4	22.7	-32.1	23.5	-34.4	24.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-2.7	-1.7	3.2	7.7	12.8	18.2	23.4	22.2	16.5	9.0	2.6	-2.6	(6)	
	DBStd	10.81	7.40	7.74	6.99	5.26	4.64	4.08	3.29	3.51	5.06	5.48	6.79	7.18	(7)	
	HDD10.0	1783	393	329	221	102	24	0	0	9	81	231	391	391	(8)	
	HDD18.3	3748	652	561	469	319	178	51	3	10	91	291	473	649	(9)	
	CDD10.0	1460	1	2	10	33	112	247	415	378	204	51	8	0	(10)	
	CDD18.3	382	0	0	0	1	8	46	160	129	36	2	0	0	(11)	
	CDH23.3	4569	0	0	2	20	124	546	1881	1501	466	31	0	0	(12)	
	CDH26.7	2010	0	0	0	2	27	198	921	686	172	6	0	0	(13)	
(14) Wind	WSAvg	4.8	6.0	5.4	4.8	4.7	4.4	4.2	3.9	3.9	4.1	4.7	5.3	5.7	(14)	
(15) Precipitation	PrecAvg	363	17	14	25	43	63	52	26	24	34	31	17	17	(15)	
	PrecMax	681	60	52	68	111	242	160	129	88	109	96	59	51	(16)	
	PrecMin	181	1	0	3	1	6	6	0	0	0	0	0	2	(17)	
	PrecStd	90	13	11	15	25	45	38	22	20	25	21	12	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	16.6	21.6	26.0	29.7	34.8	37.7	36.6	33.7	27.0	19.8	13.2	(19)	
		MCWB	5.2	6.6	9.1	11.8	14.2	16.9	17.8	17.3	15.9	13.4	8.9	5.5	(20)	
	2%	DB	10.0	12.9	18.6	22.4	26.6	31.4	35.5	34.4	31.1	23.1	16.1	10.1	(21)	
		MCWB	3.8	5.1	7.9	10.6	13.9	17.0	17.4	16.6	15.2	11.4	7.2	3.8	(22)	
	5%	DB	8.0	10.2	15.4	19.5	24.2	28.7	33.7	32.6	28.5	20.1	13.7	8.2	(23)	
		MCWB	2.7	3.6	6.7	9.3	13.2	16.1	16.9	16.2	14.2	10.1	6.2	2.7	(24)	
	10%	DB	6.2	8.0	12.7	16.7	21.6	26.4	31.8	30.7	25.9	17.2	11.4	6.2	(25)	
		MCWB	1.6	2.5	5.3	8.1	12.1	15.5	16.7	16.0	13.6	9.0	4.9	1.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	7.5	9.8	12.9	16.7	20.0	20.6	19.7	18.4	14.1	9.6	6.1	(27)	
		MCDB	11.7	15.9	19.8	23.4	25.5	28.2	31.3	30.5	27.9	24.9	18.1	12.2	(28)	
	2%	WB	4.1	5.4	8.3	11.2	15.1	18.2	19.4	18.4	16.6	12.2	7.8	4.2	(29)	
		MCDB	9.1	11.9	17.3	20.6	23.4	27.7	30.1	29.8	26.8	20.9	15.2	9.4	(30)	
	5%	WB	2.9	4.0	7.0	9.9	13.9	17.1	18.5	17.5	15.3	10.7	6.5	3.0	(31)	
		MCDB	7.5	9.7	14.6	18.5	22.4	26.5	29.2	29.1	26.2	18.6	13.0	7.6	(32)	
	10%	WB	1.8	2.6	5.7	8.6	12.7	16.1	17.7	16.7	14.1	9.6	5.3	1.6	(33)	
		MCDB	5.9	7.8	12.2	15.8	20.4	24.9	28.9	27.9	24.1	16.7	11.0	5.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	9.9	11.3	11.5	11.9	12.8	14.8	14.7	13.2	11.0	9.7	8.5	(35)	
		MCDBR	10.6	12.4	15.2	16.5	16.1	16.8	17.3	17.3	17.2	15.2	12.6	10.2	(36)	
	5% WB	MCWBR	6.6	7.1	7.8	7.8	7.0	6.4	5.5	6.0	6.7	7.3	6.8	6.2	(37)	
		MCWBR	10.5	12.0	14.4	15.0	14.5	14.8	15.4	15.3	15.4	13.9	12.0	9.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.248	0.254	0.269	0.299	0.321	0.329	0.344	0.351	0.316	0.284	0.256	0.243	(40)	
		taud	2.570	2.577	2.587	2.491	2.484	2.500	2.472	2.414	2.512	2.578	2.578	2.564	(41)	
	Ebn at Noon		895	953	977	964	946	936	916	897	907	899	874	864	(42)	
		Edh at Noon	60	72	83	100	105	104	106	108	89	72	59	54	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.70	2.67	3.78	4.76	5.61	6.71	7.10	5.96	4.54	2.87	1.82	1.32	(44)	
	RadStd		0.13	0.20	0.29	0.27	0.58	0.48	0.28	0.31	0.31	0.22	0.15	0.09	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page